

Research Protocol v1.3

Research Title

Digital Transformation and Its Impact on the Quality and Efficiency of Higher Education: A Case Study of the Universidad Nacional Mayor de San Marcos

1. Project Title

Project Description

This research investigates how digital transformation contributes to improving educational quality and institutional efficiency at the Universidad Nacional Mayor de San Marcos (UNMSM). Digital transformation is conceptualized as a strategic organizational capability that integrates digital governance, technological infrastructure, leadership, organizational culture, innovation in teaching and learning, and evidence-based decision-making. Rather than focusing exclusively on technology adoption, the project examines how these dimensions interact to generate measurable improvements in institutional performance.

The project is designed as a pragmatic mixed-methods case study capable of producing empirical evidence that informs institutional modernization while contributing to international research on digital transformation in public higher education.

2. Problem Statement

Universities worldwide are undergoing rapid digital transformation in response to technological advances, changing student expectations, increasing competition, and new quality assurance requirements. Digital technologies are transforming teaching, research, administration, and institutional governance. Nevertheless, technological investment alone does not guarantee improvements in educational quality or organizational efficiency.

In many public universities, digital transformation initiatives remain fragmented and are often evaluated through indicators of technology adoption rather than organizational performance. Consequently, university leaders frequently lack empirical evidence explaining whether digital transformation genuinely improves educational outcomes and institutional effectiveness.

Within the Peruvian higher education system, public universities are implementing digital modernization strategies aligned with national policies on

digital government and educational quality. However, systematic empirical studies examining the relationship between digital transformation, educational quality, and institutional efficiency remain limited. Existing research tends to emphasize descriptive analyses rather than explanatory models capable of identifying the organizational mechanisms through which digital transformation creates value.

The Universidad Nacional Mayor de San Marcos represents a highly relevant institutional context because of its historical leadership, organizational complexity, and strategic role in Peruvian higher education. Investigating digital transformation within this institution offers an opportunity to generate evidence capable of informing institutional policy while contributing to broader discussions regarding digital transformation in emerging economies.

3. Research Questions

Main Research Question

How does digital transformation influence the quality and efficiency of higher education at the Universidad Nacional Mayor de San Marcos?

Supporting Research Questions

1. Which dimensions of digital transformation exert the strongest influence on educational quality?
2. To what extent does digital transformation improve institutional efficiency across academic and administrative processes?
3. Which organizational, technological, and human capabilities determine the success of digital transformation initiatives?
4. How can empirical evidence generated by this research support strategic decision-making for sustainable digital transformation in public universities?

4. Research Objectives

General Objective

To evaluate the influence of digital transformation on educational quality and institutional efficiency at the Universidad Nacional Mayor de San Marcos.

Specific Objectives

- Measure the maturity of digital transformation within the university.
- Determine the relationship between digital transformation and educational quality.
- Assess the influence of digital transformation on institutional efficiency.

- Identify organizational capabilities that facilitate digital transformation.
- Develop an evidence-based framework for strengthening digital transformation in public higher education.

5. Literature Review Summary

Recent literature conceptualizes digital transformation as an organizational process rather than a purely technological initiative. Contemporary theoretical perspectives emphasize that successful digital transformation depends on leadership, governance, digital competencies, organizational culture, technological infrastructure, and institutional adaptability.

Research consistently reports positive associations between digital transformation and improvements in educational quality, operational efficiency, student engagement, organizational resilience, and institutional competitiveness. Nevertheless, significant gaps remain regarding the empirical evaluation of these relationships in public universities located in emerging economies.

Current studies also reveal limited integration between educational quality and institutional efficiency within comprehensive analytical frameworks. Most investigations examine these constructs independently, creating an opportunity for this research to contribute an integrated model explaining how digital transformation simultaneously influences both dimensions.

6. Quasi-Experimental Design

The study employs a **Mixed-Methods Case Study** supported by a quasi-experimental explanatory design.

Academic and administrative units exhibiting different levels of digital transformation will be compared using naturally occurring institutional conditions. Since random assignment is neither feasible nor ethically appropriate, the design relies on existing organizational differences to evaluate the influence of digital transformation.

Quantitative evidence will include institutional performance indicators and standardized survey instruments. Qualitative evidence will be obtained through semi-structured interviews, documentary analysis, and institutional observations. Methodological triangulation will integrate both forms of evidence to explain not only whether digital transformation influences institutional outcomes but also the organizational mechanisms responsible for those changes.

7. Data Management Plan Outline

The research adopts a comprehensive Data Management Plan consistent with FAIR principles.

Primary data will be collected through questionnaires, interviews, and focus groups involving academic staff, administrative personnel, university managers, and students. Secondary data will include strategic plans, institutional reports, accreditation documents, quality indicators, and digital transformation policies.

Data quality will be ensured through validation procedures, standardized coding protocols, secure digital storage, encrypted backups, and metadata documentation. Identifiable information will be accessible only to the research team, while anonymized datasets will be preserved to facilitate future scientific reuse in accordance with institutional regulations.

8. Ethical Considerations

The project will comply with internationally recognized ethical principles governing human-subject research.

Participation will be voluntary and based on informed consent. Participants will receive clear information regarding research objectives, procedures, confidentiality, risks, and expected benefits.

Special attention will be given to protecting institutional information and minimizing potential organizational risks arising from performance evaluation. The study will obtain authorization from the university and approval from the corresponding Research Ethics Committee before commencing data collection.

9. Project Schedule / Timeline

The project will be completed over a twelve-month period.

- **Months 1–2:** Comprehensive literature review and protocol refinement.
- **Month 3:** Instrument development and expert validation.
- **Month 4:** Pilot testing and methodological adjustments.
- **Months 5–6:** Quantitative and qualitative data collection.
- **Months 7–8:** Statistical analysis, qualitative coding, and methodological triangulation.
- **Month 9:** Integration and interpretation of findings.
- **Months 10–11:** Preparation of thesis chapters and scientific manuscripts.

- **Month 12:** Final revision, submission, and defense preparation.

10. Expected Results and Impact

- The project is expected to generate robust empirical evidence demonstrating how digital transformation contributes to educational quality and institutional efficiency within a leading public university in Peru.
- From a **theoretical perspective**, the study will contribute an integrated conceptual framework linking digital transformation, educational quality, organizational capabilities, and institutional efficiency.
- From a **methodological perspective**, the research will validate a pragmatic mixed-methods framework suitable for investigating complex organizational phenomena in higher education.
- From a **practical perspective**, the findings will support evidence-based decision-making by providing strategic recommendations for strengthening digital governance, optimizing institutional processes, and improving educational outcomes.
- Finally, from a **policy perspective**, the proposed framework is expected to be transferable to other public universities in Peru and Latin America, contributing to the development of sustainable digital transformation strategies and more effective higher education systems.